

Muhammad Shakeel

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SUMMARY

Ph.D. Candidate in Physical Chemistry studying molecular vibrational dynamics with ultrafast dual-frequency 2D IR spectroscopy, across surface-enhanced 2D IR of interfacial layers, polariton-mediated energy transfer under vibrational strong coupling, and peptide–lipid bilayer interactions probed with site-specific IR labels. Currently extending this work upstream into device fabrication, training in cleanroom photolithography and thin-film deposition to produce plasmonic and photonic substrates in house.

SKILLS

Fabrication & Prototyping: Contact mask photolithography (Suss MJB4) and e-beam thin-film deposition (Angstrom Engineering) in training, Fall 2026; CNC milling, lathe, 3D printing, PCB design, Arduino, soldering, precision measurement, hand-tool proficiency.

Optics & Instrumentation: Optical design and simulation, ultrafast laser systems, optical alignment, power and beam diagnostics, hardware troubleshooting.

Computational & Software: MATLAB, Gaussian 09, Fusion 360, Origin, Linux/CentOS cluster administration, Microsoft Office.

Analytical: FTIR and 2D IR spectroscopy, Raman and surface-enhanced Raman spectroscopy, chemometrics (PCA, PLS-DA, PLSR).

EDUCATION

Ph.D. in Physical Chemistry , Tulane University, New Orleans, LA	Exp. 2028
M.Phil. in Chemistry , University of Agriculture Faisalabad, Pakistan	2022
M.Sc. in Chemistry , University of Agriculture Faisalabad, Pakistan	2020
B.Sc. in Basic Sciences (Chemistry, Zoology, Botany) , University of Sargodha, Pakistan	2018

EXPERIENCE

Graduate Teaching Assistant, Tulane University, New Orleans, LA Aug 2023 – Present

- Instruct a cohort of ~200 students in General Chemistry Laboratory; hold weekly office hours supporting problem-solving and scientific report writing.
- Assess assignments, monthly quizzes, and laboratory demonstrations, delivering clear feedback on core chemical concepts.

Graduate Research Assistant, Rubtsov Lab, Tulane University, New Orleans, LA May 2024 – Aug 2026

- Summer research appointments (2024, 2025, 2026) in ultrafast dual-frequency 2D IR spectroscopy, covering instrument alignment and maintenance, sample preparation, data acquisition, and MATLAB-based analysis.
- Built and maintained the group's computational infrastructure and contributed to seven peer-reviewed publications.

RESEARCH PROJECTS

Tulane University

- **Nanofabrication for Polariton Optics (2026–present):** Fabricating aluminum microdisk surface-lattice-resonance (SLR) arrays on CaF₂ substrates at Tulane's Micro/Nanofabrication Facility, using contact mask photolithography and electron-beam evaporation, to bring production of vibrational strong coupling substrates in house.
- **Vibrational Polariton:** Investigated intramolecular vibrational relaxation under vibrational strong coupling using ultrafast dual-frequency 2D IR spectroscopy, revealing polariton-mediated energy transfer pathways beyond conventional molecular dynamics. Characterized fabricated photonic microcavity structures produced by collaborators — measuring mode dispersion and coupling behavior, and translating spectroscopic requirements back into device design constraints.
- **Antimicrobial Peptide–Lipid Bilayer Interactions:** Explored antimicrobial peptide–lipid bilayer interactions by probing Melp5 effects on POPC membranes with site-specific IR labels across varying temperatures, resolving peptide-induced structural and dynamical changes.
- **Linux Cluster Management for Scientific Computing:** Led setup and maintenance of a CentOS cluster for density functional theory (DFT) calculations and MATLAB simulations, improving computational throughput for the group.

University of Agriculture Faisalabad, Pakistan

- **Surface-Enhanced Raman Spectroscopy for Biofilm Detection:** Applied SERS with silver nanoparticles combined with PCA and PLS-DA to differentiate biofilm-forming bacterial strains with high sensitivity and specificity.

PUBLICATIONS

1. Cohn, B., Sribnyi, A., Nawagamuwage, S. U., Zhou, J., **Shakeel, M.**, Rubtsov, I. V., & Chuntanov, L. (2025). Controlling Intramolecular Vibrational Redistribution with an Infrared Photonic Cavity. *The Journal of Physical Chemistry Letters*, 16, 6551–6558.
2. Hussain, M., Salfi, A. B., Nawaz, H., Majeed, M. I., Rashid, N., Alshammari, A., ... & **Shakeel, M.** (2025). Screening of Blood Serum Samples With Elevated Creatinine Levels From Renal Failure Patients Employing SERS Along With Multivariate Data Analysis Techniques. *Journal of Raman Spectroscopy*.
3. **Shakeel, M.**, Majeed, M. I., Nawaz, H., Rashid, N., Ali, A., Haque, A., ... & Saleem, M. (2022). Surface-enhanced Raman spectroscopy for the characterization of pellets of biofilm forming bacterial strains of *Staphylococcus epidermidis*. *Photodiagnosis and Photodynamic Therapy*, 40, 103145.
4. Dastgir, G., Majeed, M. I., Nawaz, H., Rashid, N., Raza, A., Ali, M. Z., ... & Abdulraheem, A. (2022). Surface-enhanced Raman spectroscopy of polymerase chain reaction (PCR) products of rifampin-resistant and susceptible tuberculosis patients. *Photodiagnosis and Photodynamic Therapy*, 38, 102758.
5. Akram, M., Majeed, M. I., Nawaz, H., Rashid, N., Javed, M. R., Ali, M. Z., ... & Shahid, M. (2022). Surface-enhanced Raman spectroscopy for characterization of filtrate portions of blood serum samples of typhoid patients. *Photodiagnosis and Photodynamic Therapy*, 40, 103199.
6. Nawaz, M. Z., Nawaz, H., Majeed, M. I., Rashid, N., Javed, M. R., Naz, S., ... & Amin, I. (2023). Comparison of surface-enhanced Raman spectral data sets of filtrate portions of serum samples of hepatitis B and hepatitis C infected patients obtained by centrifugal filtration. *Photodiagnosis and Photodynamic Therapy*, 42, 103532.
7. Ali, Z., Nawaz, H., Majeed, M. I., Rashid, N., Mohsin, M., Raza, A., ... & ul Hasan, H. M. (2024). Determination of florfenicol by Raman spectroscopy with principal component analysis (PCA) and partial least squares regression (PLSR). *Analytical Letters*, 57(1), 30–40.

POSTERS & CONFERENCES

- **Gordon Research Conference on Vibrational Spectroscopy**, Southern New Hampshire University, Manchester, NH. (Poster Presentation) 2026
- **Tulane Research, Innovation, and Creativity Summit (TRICS)**, (Poster Presentation) 2026
- **Young Investigator Symposium**, Department of Chemistry, School of Science and Engineering, Tulane University Poster Presentation. 2026
- **Young Investigator Symposium**, Department of Chemistry, School of Science and Engineering, Tulane University Poster Presentation. 2025
- **American Chemical Society (ACS) Spring 2024**, New Orleans, LA 2024

HONORS & AWARDS

- Dean's Travel Award, Tulane University 2026
- Jonassen Award in Chemistry, Tulane University 2024